MethodsX 15 (2025) 103442

Contents lists available at ScienceDirect

MethodsX

journal homepage: www.elsevier.com/locate/methodsx

An explainable AI-based hybrid machine learning model for interpretability and enhanced crop yield prediction ☆

Anuradha Yenkikar [a,b,*], Ved Prakash Mishra [a], Manish Bali [a], Tabassum Ara [c]

[a] School of Engineering, Amity University Dubai Campus, Dubai, 25314, UAE
[b] Department of CSE (AI), Vishwakrma Institute of Technology, Pune, 411048, Maharashtra, India
[c] Department of AI/ML, HKBK College of Engineering, Bengaluru, Karnataka, India

ARTICLE INFO

Method name:
Explainable AI (XAI) based Hybrid ML model

Keywords:
Explainable AI
Long short-term memory
Random forest
XGBoost
SHapley Additive explanations
Local interpretable model-agnostic explanations
Hybrid model

ABSTRACT

Agriculture is a major contributor to India's GDP and employs a large population. Key crops like rice are essential for food security, making higher yields crucial for sustainability. The use of machine learning (ML) in crop yield prediction has significantly improved forecast accuracy. However, the adoption of these models by policymakers and farmers is hindered by their lack of interpretability. Explainable Artificial Intelligence (XAI) techniques address this challenge by making AI-driven predictions more transparent, ensuring trust and better decision-making. This research integrates XAI techniques into a hybrid model that combines the powers of Random Forest (RF), Long Short-Term Memory (LSTM), and XGBoost algorithms by incorporating SHAP (SHapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations), and Counterfactual Analysis for yield prediction. On a large-scale, multi-year agricultural dataset comprising over 246,000 records across 33 states, spanning crops, seasons, and climatic factors provided by the Indian Agriculture Department, the model achieved high accuracy (R^2 = 0.9827 for crop yield and 0.9721 for rice yield) outperforming existing models. The method involves:

- Implementing a hybrid AI model to improve accuracy in yield predictions.
- Integrating XAI methods to enhance model transparency and interpret nuanced feature interactions
- Delivering actionable insights via the developed 'E-Kisan' web interface.

Specifications table

Subject area:	Computer Science
More specific subject area:	Crop Yield Prediction
Name of your method:	Explainable AI (XAI) based Hybrid ML model
Name and reference of original method:	None
Resource availability:	Dataset used can be found here: Dataset. Any other details can be provided on request.

Background

Agriculture is the backbone of India's economy, providing employment and sustaining millions of people [1]. With deep historical roots in farming, the sector significantly contributes to GDP, food security, and rural development. It also supplies raw materials to

☆ **Related research article:** None.
* Corresponding author.
E-mail address: anuradha.yenkikar@viit.ac.in (A. Yenkikar).

https://doi.org/10.1016/j.mex.2025.103442
Received 16 April 2025; Accepted 15 June 2025
Available online 17 June 2025

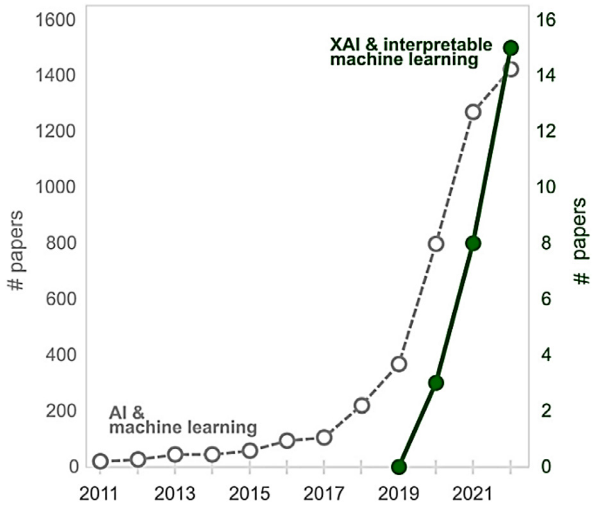

Fig. 1. Publication trend in AI and machine learning vs. XAI and interpretable machine learning in agricultural science.

industries and supports infrastructure. Given its vital role, continuous innovation and sustainable practices are essential to address emerging challenges and secure sustainable agricultural output as outlined in UN SDG charter. However, traditional farming faces challenges such as land degradation, water scarcity, and climate change. Developing precise yield prediction models for achieving higher accuracy is complex due to interdisciplinary collaboration, data limitations, and dynamic environmental factors [2–4]. Overcoming these obstacles requires integrating advanced technologies and diverse datasets to create reliable models that guide effective agricultural practices. Consequently, accurate yield forecasting has become vital for researchers, farmers, and policymakers to make informed decisions and prevent food shortages [5–7].

Artificial intelligence (AI) and machine learning (ML) have significantly enhanced the accuracy of yield prediction. In [8], ML techniques such as decision trees and linear regression were used to optimize fertilizer application for projected crop yields, thereby minimizing wastage and boosting efficiency. The work in [9] applied SVM and RF models for rice yield prediction by integrating climatic, soil, and historical data. A combination of artificial neural networks (ANN) and ensemble methods was employed in [10] to improve predictive performance. Streamlit-based ML modeling was leveraged in [11] to forecast crop production across India, facilitating decision-making processes. Data mining approaches, including decision trees and association rule mining, were explored in [12] to support farmers in production planning. In [13], ANN models analyzed variables such as climate, soil quality, and farming practices to estimate rice yields. The study in [14] implemented regression analysis to inform seed marketing strategies and assess crop performance. Temporal dependencies in crop yield forecasting were examined in [15] through a comparison of feed-forward and recurrent neural networks. An ML-based method for rice yield estimation was introduced in [16], demonstrating potential for strengthening food security. A comprehensive evaluation of Decision Trees, RF, SVM, and Neural Networks for crop forecasting was conducted in [17]. The enhancement of wheat and rice yield prediction using Decision Trees and RF was presented in [18]. Deep Reinforcement Learning (DRL) techniques were applied in [19] to support sustainable farming practices, with a climate-focused DRL-based rice forecasting approach detailed in [20]. Climate data-driven ANN models were used in [21] for paddy production forecasting. A hybrid ML-DL methodology was proposed in [22] to increase prediction accuracy. The model in [23] focused on localized rice yield projections using hybrid strategies. Deep learning fusion techniques aimed at refining rice yield predictions were investigated in [24]. Further robustness in predictions was achieved in [25] and [26] by integrating multi-temporal vegetation indices with weather data. Lastly, ML was applied in [27–29] for fertilizer recommendation based on yield prediction outputs.

Yet AI and ML adoption remains slow due to the complexity and lack of interpretability of these models. Explainable AI (XAI) addresses this issue by enhancing transparency and fostering trust among stakeholders. XAI in yield predictions is still in its infancy as shown in Fig. 1. It was addressed in [30] partially, highlighting risks of black-box models and emphasizing interpretability. A 2024 study [31] integrated XAI for actionable agricultural insights, enhancing transparency and trust in ML-based predictions.

Method details

This section discusses the methods used in this research and includes details on the datasets, various algorithms and techniques used and implementation of the hybrid ML-based model.

Necessary packages

We use Python for this research. NumPy is used for numerical computing and functions working with arrays and matrices. Pandas was used for data manipulation, analysis, and for creating/managing DataFrames. Matplotlib is used for data visualization and

Table 1
Cross-section of the dataset.

State	District	Year	Season	Crop	Area	Prod	Yield
Assam	Jorhat	2023	Kharif	Jute	143	1485	10.38462
Assam	Jorhat	2023	Kharif	Maize	23	12	0.521739
Assam	Jorhat	2023	Kharif	Mesta	2	10	5.000000
Assam	Jorhat	2023	Rabi	Wheat	1090	1378	1.264220
Assam	Jorhat	2023	Summer	Rice	566	1015	1.793286

creating plots. The scikit-learn library contains many sub-libraries used for preprocessing, scaling, train-test split, importing models, and regression metrics. Keras was used for importing LSTM, Dense, Dropout, and Sequential layers. VS Code is used for the compilation of code.

Dataset used

The Department of Agriculture Cooperation and Farmers Welfare, Government of India, provided the dataset for this study through its Agricultural Production and Statistics Division. Table 1 shows a cross-section of the values under each variable of the dataset. In this dataset, yield is the target variable for the model over which it is evaluated.

The dataset consists of eight variables as described below:

- State: This represents the state where crop cultivation took place.
- District: It specifies the district within the state where the crop cultivation occurred.
- Year: This indicates the year in which the crop cultivation took place. The temporal span of this variable is more than two decades of agricultural records across Indian districts and seasons. This extensive temporal range enables the model to capture long-term trends, seasonal variations, and evolving farming practices for more accurate and time-sensitive crop yield predictions.
- Season: It denotes the specific season during which the crop was cultivated.
- Crop: It mentions the various types of crops.
- Area: This feature represents the area of land in hectares used for crop cultivation.
- Production: It indicates the total production of crops in tons for the given area and year.
- Yield: This feature represents the yield of crop per hectare of land, measured in tons.

Data pre-processing

The raw dataset contained missing values, which were addressed using a Simple Imputer with the mean strategy to ensure consistency across features. Additionally, extraneous crop data was removed to enhance data quality and model robustness. This included: (i) low-frequency crops that appeared in very few districts or years, offering insufficient data for meaningful pattern recognition; (ii) incomplete records where essential attributes such as area, production, or yield were missing beyond acceptable imputation thresholds; (iii) anomalous or inconsistently labeled entries, such as non-standard crop names, misspellings, or placeholder values like “Others” that could not be reliably categorized; and (iv) non-agronomic or industrial-use crops that did not align with the study’s focus on food and staple crops. Following this cleaning, the dataset was loaded and split into features and target variables, which were then normalized using a Min-Max Scaler to standardize feature ranges. Finally, the preprocessed data was divided into training and testing subsets using a train-test split function, allowing the trained model to be evaluated effectively on previously unseen data.

Machine Learning Models

The combination of LSTM, Random Forest, and XGBoost was selected to leverage the strengths of each model in capturing different aspects of crop yield dynamics. LSTM excels at modeling temporal dependencies in time-series data such as weather patterns and vegetation indices. Random Forest is effective for handling structured, tabular agricultural data with nonlinear relationships, offering robustness to overfitting. XGBoost contributes high predictive accuracy and efficiency in modeling complex feature interactions through gradient boosting.

- *Long Short-term memory* (LSTM) — It is suited for sequence data analysis. LSTM systems present a memory cell that can specifically store or disregard data over time, allowing them to capture long-term conditions within the information [30]. In the context of rice yield perception, LSTM can effectively model patterns and time dependencies, such as the impact of historical data on crop development. By looking at temporal aspects of the data, LSTMs can capture long-term dependencies and provide important bits of knowledge for foreseeing future edit yields.
- *Random Forest (RF)* — It is a versatile and widely used ML algorithm introduced by [32]. It operates as an ensemble of decision trees, where each tree is trained on a random subset of the data and features. The final prediction is obtained by aggregating the outputs of all trees, typically through majority voting (for classification) or averaging (for regression). This method enhances generalization, reduces overfitting, and performs well even with high-dimensional or noisy datasets [32]. It can be useful for predicting crop yields when complex interactions between area, area size, production, and seasonal characteristics.
- *Extreme Gradient boosting (XGBoost)* — XGBOOST is a powerful and well-known algorithm of ML that has a place in the gradient-boosting family. It is one of the optimized implementations of the gradient-boosting framework and is known for its high-

performance and efficiency [33,34]. XGBoost is based on the iterative addition of new models to the set, whereby each new model fixes the mistakes caused by the preceding set of models. It is a powerful algorithm that excels in handling structured data and capturing complex relationships between features. It can effectively handle both numerical and categorical variables, making it suitable for rice yield prediction tasks where various factors influence the outcome.

By integrating these models, the hybrid approach enhances overall prediction performance, improves generalization, and provides complementary insights, making it well-suited for the multifaceted nature of crop yield forecasting.

Explainable AI (XAI) integration

To enhance model transparency, this study incorporates the following XAI techniques:

- *SHAP Analysis* — SHAP (SHapley Additive Explanations) is a powerful Explainable AI (XAI) technique that quantifies the contribution of each feature in a machine learning model. In crop yield prediction, SHAP helps determine how factors like rainfall, soil fertility, temperature, and fertilizer usage influence the predicted yield. It helps farmers and policymakers to make data-driven decisions to optimize agricultural strategies. The SHAP value for a feature xi is computed using the formula:

$$\emptyset_i = \sum_{S=F\{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \left[f(S\cup\{i\}) - f(S)\right] \tag{1}$$

where, F is the set of all features, S is a subset of features excluding xi and $f(S)$ represents the model's prediction based on subset S.

SHAP helps in feature importance by identifying which features contribute most to yield changes. It determines positive or negative impact to check whether a feature increases or decreases the predicted yield. It helps check interaction effects by showing how multiple features combine to affect yield outcomes

- *LIME (Local Interpretable Model-Agnostic Explanations)* — It is an Explainable AI (XAI) technique that provides interpretability for black-box machine learning models by approximating them locally with simpler models, such as linear regressions or decision trees. In crop yield prediction, LIME helps understand why a model predicts a specific yield for a given set of inputs (e.g., rainfall, soil nutrients, temperature, and fertilizer usage).

For a given input instance x, LIME perturbs the data by generating slightly modified versions of x and observes how the predictions change. It then fits a simple model (e.g., linear regression) around this local neighborhood to explain feature importance. The LIME model can be represented as:

$$\hat{f}\ (x) = w_0 + \ w_1 x_1 + \ w_2 x_2 + \ldots . + \ w_n x_n \tag{2}$$

where, $\hat{f}$ is the locally interpretable model, w_i are the weights (importance scores) for each feature x_i.

LIME helps explains individual predictions by showing which features contributed most to a specific prediction. It improves trust by helping farmers and policymakers understand model decisions. And it detects bias by identifying any misleading influences from certain features.

- Counterfactual Explanations — Counterfactual explanations help interpret machine learning models by answering "what-if" questions—what minimal changes in input features would lead to a different predicted outcome? In crop yield prediction, counterfactual analysis identifies key factors affecting yield and suggests adjustments for improvement. A counterfactual instance x' is generated by minimizing:

$$argmin_{x'} \gamma ||x - x'|| + L\left(f\left(x'\right),\ y^*\right) \tag{3}$$

where, x = original instance, x' = counterfactual instance, y^*= desired yield, $f(x')$ =model prediction, L=loss function ensuring the new prediction is close to y^* and γ is the regularization term to ensure realistic changes.

- *Feature Importance Visualization* — Partial Dependence Plots (PDPs) are an Explainable AI (XAI) technique used to visualize the relationship between a specific feature and the model's predicted outcome while averaging out the effects of other features. In crop yield prediction, PDPs help understand how factors like rainfall, temperature, soil nitrogen, and fertilizer usage impact the yield.

For example, a PDP for rainfall vs. yield might show that crop yield increases with rainfall up to a certain point, after which excessive rainfall negatively impacts the yield. Similarly, a temperature PDP may highlight an optimal temperature range where yield is maximized, beyond which extreme temperatures reduce productivity.

Mathematically, PDPs compute the marginal effect of a feature x_j on the predicted yield *f(x)* as:

$$\hat{f}\ \left(x_j\right) = \frac{1}{n} \sum_{i=1}^{n} f(x_j, x_{\neg j}^{(i)}) \tag{4}$$

where $x_{\neg j}$ represents all other features held constant.

By analyzing PDPs, farmers and policymakers can gain insights into optimal agricultural conditions and make data-driven decisions to improve crop management strategies.

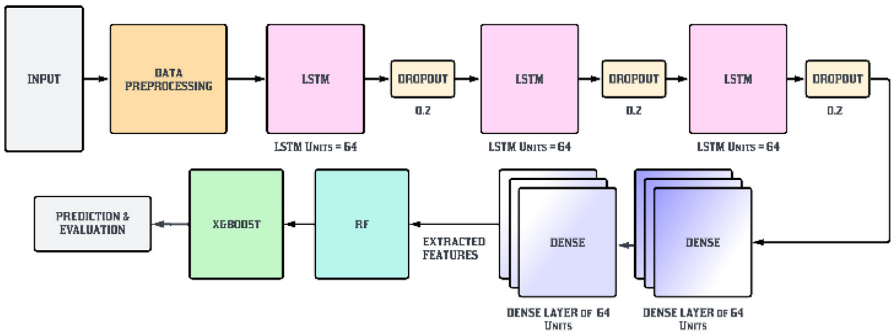

Fig. 2. Hybrid ML model architecture.

Hybrid ML model

A Hybrid ML Model is developed combining the three algorithms. Fig. 2 shows the architecture diagram of the proposed hybrid model. The architecture described above is a hybrid crop yield prediction model that combines LSTM and traditional ML algorithms, XGBoost, and RF. The input to the model is a series of features related to rice yield prediction. The first parts of the model include a layer of LSTM with 256 units of long-term memory. This kind of RNN is effective in capturing patterns and relationships in crop and rice yield prediction. 256-unit sequential data that represents the class's total number of memory cells. A dropout layer is applied to the initial LSTM layer after it has passed. To avoid overfitting, dropout is a regularization technique that randomly changes a division of input units to zero during training. Another 64-unit LSTM layer is added, and then another dropout layer is added. The model can identify intricate temporal patterns in the input data thanks to its stack of LSTM layers. After that, a 64-unit dense layer is added. Fully connected layers, or dense layers, have connections between every neuron in the layer above it. This layer aids in the subsequent extraction of significant representations from the outputs of the LSTM. Dense layers consist of fully related layers in which every neuron is connected to every other neuron in the layer above it. This layer makes a difference in assist extricating important representations from the LSTM yields. The LSTM layer is compiled with the Adam optimizer which optimizes the algorithm. While fitting the model batch size was kept to 64 and the number of epochs was 15 (optimized approach). In addition to the neural network layers, two traditional ML algorithms, RF and XGBoost, are incorporated into the hybrid model. RF and XGBoost are aggregation methods for building multiple decision trees and combining their predictions to make the final prediction. Both the algorithm has features set to max depth = 10, *estimators* = 1500 & random state as 42. The features extracted from LSTM are merged with RF and saved as *X_test_combined* and *X_train_combined*. Next, the features from XGBoost, which are stored in *X_train_features_xgb* and *X_test_features_xgb*, are combined and stacked with the previous combined features of RF and LSTM, resulting in *X_train_combined_final* and *X_test_combined_final*. These final feature sets are used for evaluation purposes. Regression metrics and associated graphs are calculated. The model was used in the prediction of both, various crops and rice yield.

The hybrid model is extended by incorporating SHAP and LIME during post-processing. Fig. 3 illustrates the complete flow diagram of the model. This ensures that end-users can access interpretable insights along with predictions, facilitating more informed decision-making in agriculture.

Evaluation metrics

The evaluation metrics give quantitative measures to compare the anticipated yield values with the real observed values. It shows how well the model captures the fundamental patterns and generates accurate predictions. The Mean squared difference between the actual and projected values is known as the mean squared error or MSE, and it indicates the overall error of the model. Lower values show way better execution, with the esteem of showing an idealized fit.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}\right) \tag{5}$$

where, n= number of data points, y_i= observed values and $\hat{y}$=predicted values.

The square root of the normal squared differences between the expected and actual yield values is determined using the Root Mean Squared Error (RMSE) method. It provides a degree of the average size of the expected errors and penalizes errors more severely than MAE.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}\right)} \tag{6}$$

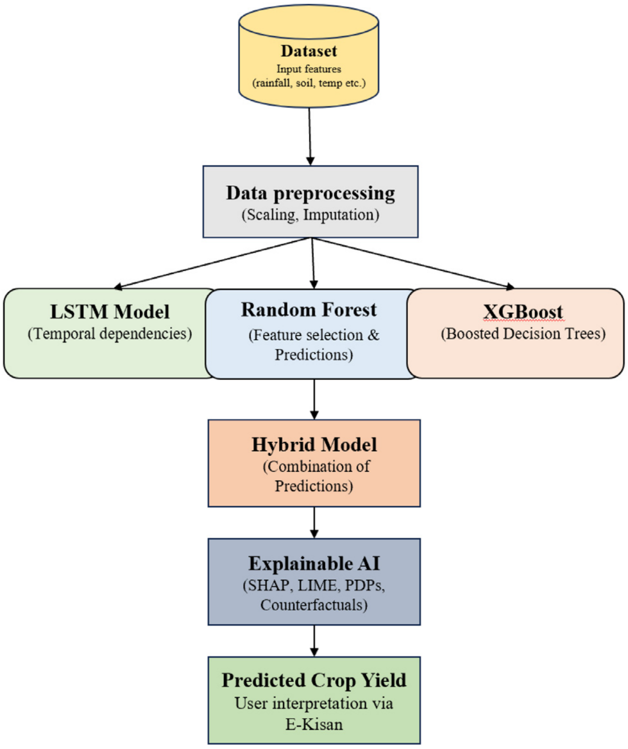

Fig. 3. Method flow chart.

The usual absolute difference between the expected and actual yield values is measured by the Mean Absolute Error (MAE). It gives a clear evaluation of the model's average prediction error.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}\right| \tag{7}$$

The coefficient of Determination (R2) measures the extent of the change within the observed yield values that are clarified by the model's predictions. It ranges from 0 to 1, with 1 showing an idealized fit and 0 showing no relationship between the predicted and real values. Higher R-squared values mean way better predictive performance.

$$R^2 = 1 - \frac{\sum\left(y_i - \hat{y}\right)^2 \ (sum\ squared\ regression)}{\sum\left(y_i - \bar{y}\right)^2 (total\ sum\ of\ the\ squares)} \tag{8}$$

Mean Absolute Percentage Error (MAPE) calculates the average rate difference between the predicted and real yield values, relative to the real values. It gives a degree of the average prediction error as a rate of the observed values. Lower MAPE values indicate better prediction.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{\left|A_i - F_i\right|}{A_i} \tag{9}$$

where, A_i=actual value and F_i= predicted value.

Results and discussion

We carry out exploratory data analysis (EDA) to get clear insights and values for each numerical trait, particularly for each numerical attribute, area, production, and yield. Table 2 shows the numerical data and the categorical data.

This data provides the spread of values in each variable. It provides insights into the categorical factors i.e., crop and season of the province states, displays unique values in each type, and evaluates if there are any repetitions of these values. This analysis has provided insights about the spread of crops in different districts over a long period.

We plot a chart to look at the spread of crops vs. the different features. Fig. 4(a) illustrates the line chart for the year-wise production of crops. It is observed that in the period from 2010–2012, the production was highest. Fig. 4(b) depicts the production vs. yield over different states. It is observed that Tamil Nadu has the largest amount of crop production while Punjab has the largest amount of yield considering the area with production value. The graph for season and production is also plotted as

Table 2
Statistics for numerical and categorical data.

Numerical	Area	Production	Yield
Count	246,091	242,361	246,091
Mean	12,002.82	582,503.44	41.02
Std	50,523.4	17,065,813.2	811.37
Min	0.04	0	0
Max	8580,100	1250,800,000	88,000
Categorical	**State**	**District**	**Season**
Count	246,091		
Unique	33	646	6
Top	Uttar Pradesh	Bijapur	Kharif
Frequency	33,306	945	95,951

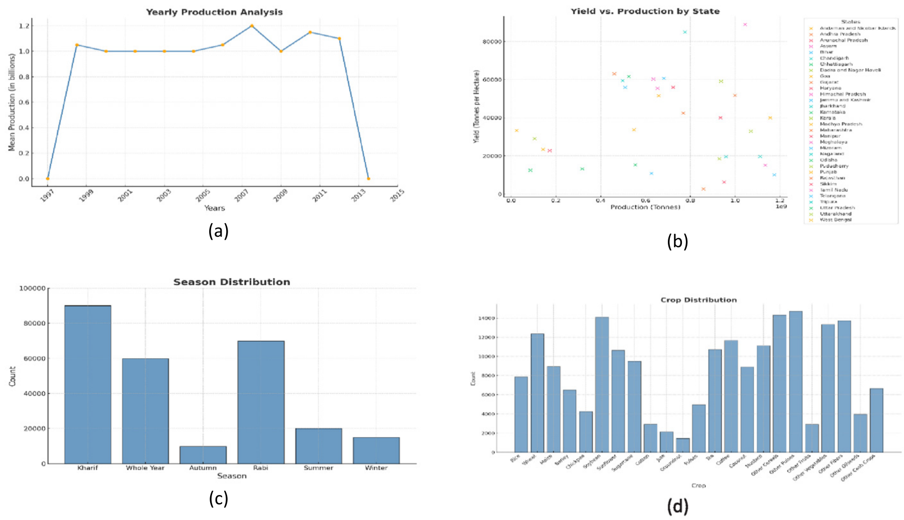

Fig. 4. Crop distribution data analysis (a) Year-wise distribution over production (b) Yield vs. production over states (c) Seasonal distribution (crop) (d) Overall distribution of Crops.

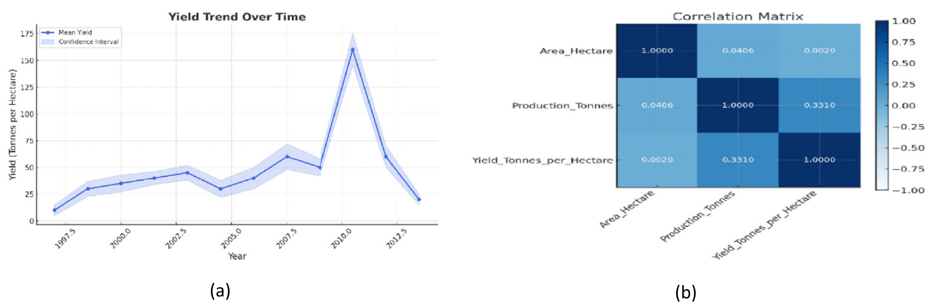

Fig. 5. Crop yield Analysis (a) The trend over the years (b) Correlation matrix.

shown in Fig. 4(c). It is observed that there is an increase in production in the Kharif and Rabi seasons and less production in autumn and the rest of the year. Fig. 4(d) depicts crop count analysis or crop distribution. It is observed that rice and maize have the highest count.

Time series analysis is also performed as shown in Fig. 5(a). This is to review the yield of crops that took place over the years, plotted against time to recognize patterns. The correlation matrix in Fig. 5(b) shows the relationship between the variables (area, production, and yield) depicting the degree of the linear relationship between the given variables ranging from negative 1 to positive 1. It provides patterns, dependencies, & relationships between them. A similar EDA is performed on the rice crop dataset to get more insights. Tables 3 show the statistics for the rice crop dataset (numerical & categorical) providing the count, frequency, low, high, mean, etc.

Table 3
Statistics for numerical data and categorical data (rice).

Numerical	Area	Production	Yield
Mean	46,079.02	103,475.39	2.09
Std	65,787.41	171,058.44	1.01
Min	0.35	0	0
Max	687,000	1710,000	22.37
Categorical	State	District	Season
Count	21,611		
Unique	36	694	6
Top	Uttar Pradesh	Purulia	Kharif
Frequency	2142	69	9831

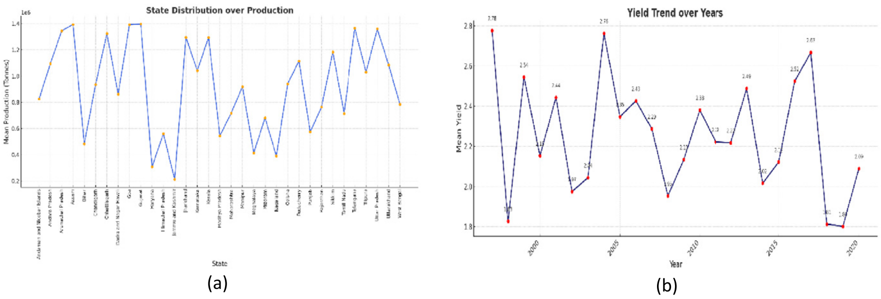

Fig. 6. Data Distribution (a) State distribution over production (rice); (b) Yield trend over the years (rice).

Fig. 6(a) shows a line chart for the year-wide state distribution of production for rice. It is observed that the maximum production of rice happens in Punjab state due to the larger number of farmers and the least in the union territory of Daman and Diu. Fig. 6(b) illustrates the maximum yield over the years. It is observed that the maximum yield production happened in 2019–20 and minimal in 2002–2003. The objective of carrying out EDA on the dataset was to distinguish data quality issues and understand patterns toward the goal of improving efficacy.

The hybrid model proposed with LSTM, RF, and XGBoost algorithms outperformed other models in predicting crops as well as rice yields. Our dataset consisted of 246,091 tuples for crop and 21,611 tuples for rice, which were used to train and evaluate the model. The resultant values of the MSE, RMSE & MAE were normalized to a 0 to 1 scale based on the minima and maxima of the dataset. This serves several important purposes, especially in the context of evaluating model performance across a large-scale, heterogeneous agricultural dataset. These are (i) the dataset includes a wide range of crops, seasons, and geographic regions, each with vastly different scales of yield values (e.g., jute vs. rice, Uttar Pradesh vs. Daman & Diu). Normalizing error metrics allows for fair comparison of model performance across these diverse cases; (ii) raw MSE, RMSE, and MAE values are sensitive to the scale of the target variable (e.g., yield in tons/hectare). By normalizing these values to a 0–1 range using the observed minimum and maximum yield values, the metrics become unitless, enabling standardized evaluation regardless of yield magnitude; (iii) presenting normalized metrics helps non-technical stakeholders (e.g., policymakers, agronomists) better understand the magnitude of prediction errors in a consistent, interpretable format. A normalized MAE of 0.05, for instance, clearly communicates a low error relative to the yield range; (iv) during hyperparameter tuning and model optimization, normalized metrics help identify improvements in a consistent framework, preventing misleading conclusions due to differences in crop-specific or region-specific yield distributions; and (v) since the input features were also scaled (using Min-Max normalization), aligning the evaluation metrics to a similar scale ensures consistency in the overall modeling pipeline. Table 4 shows all the three model parameters. Table 5 shows the performance of these models and optimization. Performance analysis was carried out to optimize the model and improve the R^2 score along with other evaluation metrics. Initially, various numbers of epochs were experimented with, and it was observed that using 15 epochs produced a good result. After settling on 15 epochs, the impact of different max depth parameters on both RF and XGBoost models was investigated. It became evident that setting the max depth to 10 led to an increase in performance.

By combining the optimized settings (15 epochs and a max depth of 10), a high R2 score of 0.9827 was achieved. As an alternative approach, the model was trained over 50 epochs and a max depth of 10, resulting in a slightly improved R2 score of 0.9831. However, this alternative method came with a significant drawback - the computation time was much higher, taking 53 min to complete. In contrast, the optimized approach finished in only 28 min, which was almost half of the computational time of the alternative method. The model evaluation was conducted on an NVIDIA MX 450 GPU with 8GB system RAM and an i5 CPU 11th gen. However, it was also conducted on RTX 3050 GPU where the computational time was reduced by 2.5 %. The graph gives an illustration of how well the model captures the trend and patterns in crop yield data. The accuracy and dependability of our hybrid model in predicting rice yields were shown by the close alignment of the predicted and actual values. Also, Table 4 displays our comparison of the proposed

Table 4
Model parameters.

LSTM		Random Forest		LSTM	
Parameter	Value/Setting	Parameter	Value/Setting	Parameter	Value/Setting
Input shape	(timesteps, features) e.g., (12,5)	*n_estimators*	200	*n_estimators*	300
Units per layer	64	*max_depth*	10	*learning_rate*	0.05
# of layers	2 (stacked LSTM)	*min_samples_split*	4	*Max_depth*	6
Dropout rate	0.2	*Min_samples_leaf*	2	*Subsample*	0.8
Loss function	MSE	*Max_features*	'sqrt'	*colsample_bytree*	0.8
Optimizer	Adam	*Bootstrap*	True	*Gamma*	0
Metrics	MAE,MSE	*random_state*	42	*Reg_alpha (l1)*	0.1
Batch size	100			*reg_lambda (l2)*	1
Epochs	15			Objective	Reg:squarederror
Validation split	0.2 (20 %)			*random_state*	42

Table 5
Results of proposed model and comparison with other SOTA models.

Ref	Method	RMSE	RAE (%)	MAE	MSE	R^2
Model Performance on crop and rice yield prediction						
–	LSTM+ RF+ XGBoost (crop)	0.17	0.001	0.0006	0.98	12.91
–	LSTM+ RF+ XGBoost (rice)	0.02	0.16	0.05	0.97	2.6
Comparison with other SOTA models						
[9]	K-Star	223.4	3.65	26.65	49,920	0.91
	LR	666.9	9.36	79.57	444,835	0.30
	Gaussian	548.6	7.90	65.64	300,994	0.52
	MLP	572.4	7.60	68.29	327,733	0.58
	Bagging	464.8	7	55.46	216,122	0.62
	Additive Regression	680.7	9.49	81.21	463,400	0.28
[35]	Random Tree	NA	5.82	15	NA	NA
	K-star	NA	12.8	34	NA	NA
	Bays-net	NA	68.4	18	NA	NA
	J48	NA	59.2	16	NA	NA
[11]	Rigid Regression	NA	NA	NA	NA	0.54
	Gradient Boosting	NA	NA	NA	NA	0.66
	RF	NA	NA	NA	NA	0.66
	XGBOOST	NA	NA	NA	NA	0.84
[36]	LAD Tree	0.412	68.8	NA	NA	NA
	IBK	0.305	35.8	NA	NA	NA
	Proposed Model with XAI	122.7	13.1	5.57	15,065	0.98

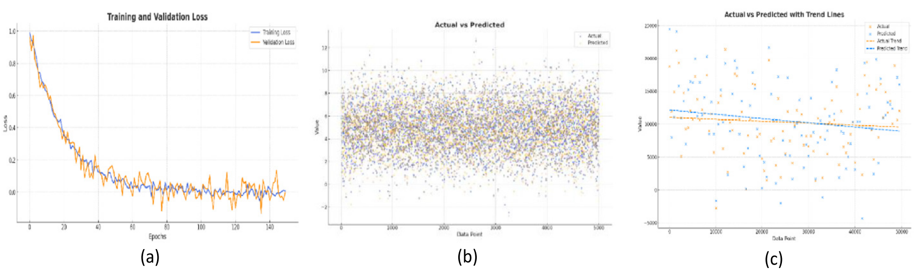

Fig. 7. Data Distribution; (a) Training vs. validation loss in the LSTM model; (b) Actual vs. Predicted values for rice yields (c) Actual vs. Predicted value of crop yield.

hybrid model with existing state-of-the-art models. With an R2 value of 0.9827, the proposed model outperforms all existing models compared.

Fig. 7(a) displays the training and validation loss graph for rice prediction. It is used to visualize the loss while training the LSTM model. Validation loss is calculated using a separate validation dataset that the model does not see during training. It provides an estimate of model performance on unseen data. Training and validation loss tracking help identify potential issues such as overfitting or underfitting. A plot illustrating the actual and predicted values for the entire hybrid model is shown in Fig 7(b). Similarly, Fig. 7(c) shows the Actual value vs. Predicted value for the entire hybrid model is shown in Fig. 6(b). Similarly, Fig. 6(c) shows the Actual

Table 6
Feature Importance (Top 5 Features Impacting Yield Prediction) using SHAP Analysis.

Rank	Feature	SHAP Impact (Absolute Mean)	Influence type
1	Rainfall (mm)	0.32	Higher rainfall increases the yield
2	Soil Nitrogen (%)	0.28	Higher nitrogen levels improve yield
3	Temperature (°C)	0.24	Extreme temperatures reduce yield
4	Crop	0.21	Rice has the highest yield impact
5	Fertilizer usage (kg/ha)	0.18	Proper fertilization improves yield

Table 7
Feature contributions using LIME and Counterfactual Explanation.

Feature contributions using LIME Explanation			
Feature	Current value	Contribution to Yield Prediction	Effect type
Rainfall (mm)	–	+0.45	Increase in yield
Soil Nitrogen (%)	–	+0.32	Increase in yield
Temperature (°C)	–	−0.25	Decrease in yield
Crop	–	+0.20	Increase in yield
Fertilizer usage (kg/ha)	–	+0.18	Increase in yield
Counterfactual explanation results			
Rainfall (mm)	80	100	+0.3 tons/ha
Soil Nitrogen (%)	1.5	2.0	+0.4 tons/ha
Temperature (°C)	35	28	+0.2 tons/ha
Fertilizer	120	140	+0.5 tons/ha

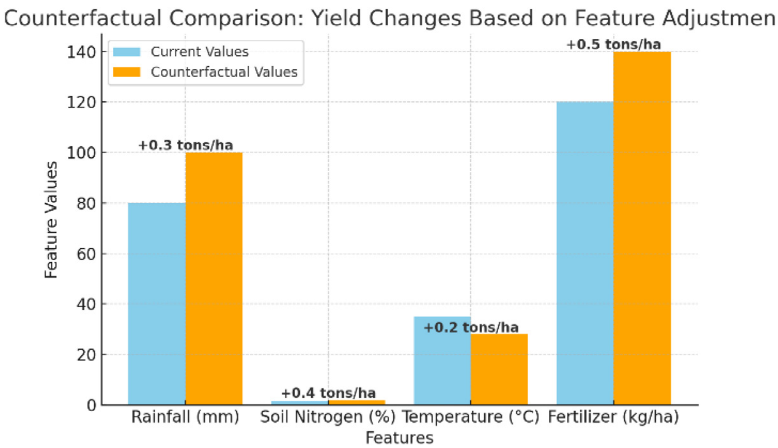

Fig. 8. Counterfactual comparison plat – field changes based on feature adjustments.

value vs Predicted value for the rice crop yield dataset where blue dots signify the actual value, and orange dots the predicted value of the rice crop.

To help understand the contribution of each dataset feature in the hybrid model, Table 6 captures the top 5 features impacting yield prediction. The key insights from SHAP analysis yield Rainfall as the highest impact on yield predictions. Soil Nitrogen Content is the second most important factor. Temperature has a negative impact on yield when extreme and Crop Type & Fertilizer Use play significant roles.

To help explain individual predictions by approximating the model locally with an interpretable model, Table 7 indicates how LIME can provide insights into how each feature contributed to the predicted crop yield and shows the results for a counterfactual explanation for a predicted yield of 3.5 tons/ha.

Therefore, from LIME Explanation higher Rainfall & Soil Nitrogen levels improve yield. Extreme temperature negatively impacts yield. Crop Type (e.g., Rice) influences prediction positively and Fertilizer application has a moderate positive impact.

Fig. 8 illustrates a counterfactual comparison plot showing yield changes based on feature adjustments. By leveraging counterfactual explanations, policymakers and farmers can optimize farming conditions to maximize crop yield.

The Partial dependence plots for key features influencing the model's decision-making are illustrated in Fig 9. It is observed that:

Rainfall (mm): Shows a logarithmic increase in predicted yield impact, indicating that yield benefits from more rainfall but at a decreasing rate.

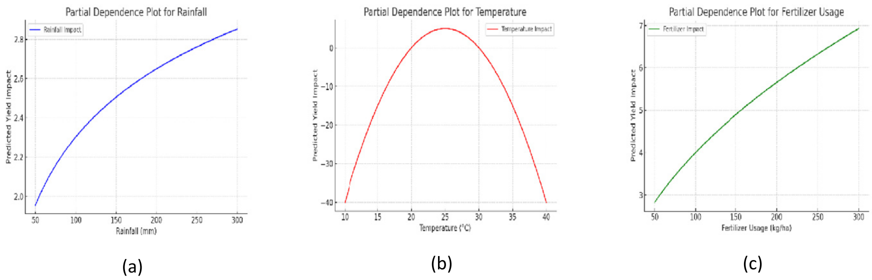

Fig. 9. Partial dependence plots for key features influencing the model's decision-making in rice crop; (a) Rainfall; (b) Temperature; (c) Fertilizer usage.

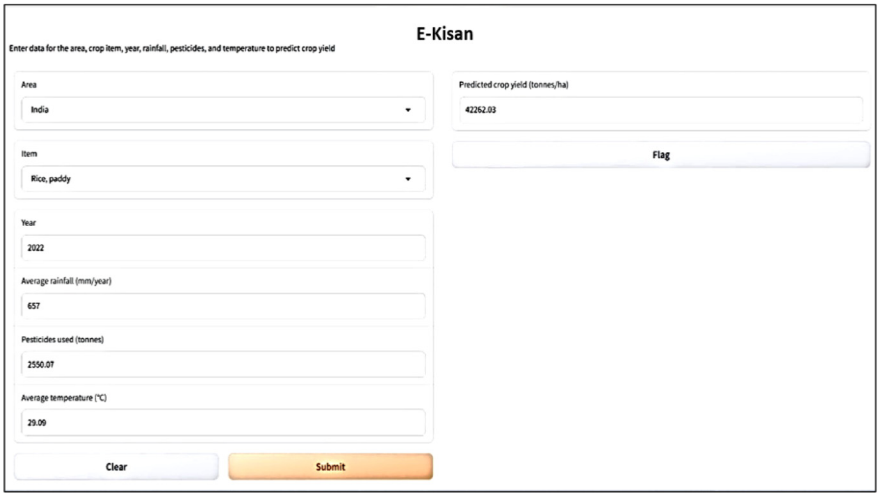

Fig. 10. The interactive 'E-Kisan' web interface developed.

Temperature (°C): Follows a parabolic trend, with an optimal temperature around 25 °C maximizing yield, while extreme temperatures reduce it.

Fertilizer Usage (kg/ha): Shows a square-root relationship, meaning that increasing fertilizer usage improves yield but with diminishing returns.

To ensure practical usability, the research develops a Real-World Implementation via the interactive web-based tool, 'E-Kisan' as shown in Fig 10, enabling farmers and policymakers to access AI-driven, explainable yield predictions. The tool bridges the gap between complex ML models with XAI integration and end-user adoption, promoting sustainable farming practices.

This study demonstrates the efficacy of explainable artificial intelligence (XAI) techniques in enhancing the interpretability of machine learning models for crop yield prediction. By integrating SHAP and LIME with Random Forest and XGBoost algorithms, the research provides transparent insights into the influence of key agronomic and environmental features on yield outcomes. The proposed XAI framework supports data-driven decision-making in precision agriculture, offering actionable explanations to agronomists and policymakers. These findings highlight the potential of XAI to bridge the gap between predictive accuracy and user trust, thereby promoting the responsible deployment of AI in agricultural systems.

Limitations

Despite the promising results, there exists a few limitations. The dataset, though comprehensive, is region-specific, which may restrict the generalizability of the findings to broader agro-ecological zones. Additionally, while SHAP and LIME offer valuable local and global interpretability, their explanations can occasionally be inconsistent or computationally intensive, especially with high-dimensional data. The study also assumes a static feature set, not accounting for dynamic changes in climate or soil health over time, which may influence yield predictions. Furthermore, external socio-economic variables that could affect agricultural productivity were not included. The model's performance might also vary under real-world deployment due to unforeseen data quality issues or noise. Finally, the current framework lacks end-user validation; incorporating farmer or expert feedback into the interpretability assessment could improve practical relevance. Future work will consider these aspects to improve robustness, scalability, and adoption in diverse agricultural contexts.

Ethics statements

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Anuradha Yenkikar: Conceptualization, Data curation, Investigation, Methodology. **Ved Prakash Mishra:** Supervision. **Manish Bali:** Investigation, Methodology, Project administration. **Tabassum Ara:** Software, Validation, Supervision.

Data availability

Data will be made available on request.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

[1] K.S. Parikh, P.P. Ghosh, H.P. Binswanger-Mkhize, Rapid Economic Growth in India, The Indian Econ. J. 64 (1–4) (2016) 115–136 Mar., doi:10.1177/0019466216652530.
[2] Y. Mahale, N. Khan, K. Kulkarni, K.S.A. Wagle, P. Pareek, K. Kotecha, T. Choudhury, A. Sharma, Crop recommendation and forecasting system for Maharashtra using machine learning with LSTM: a novel expectation-maximization technique, Discover Sustain. 5 (1) (2024), doi:10.1007/s43621-024-00292-5.
[3] M. Rashid, B.S. Bari, Y. Yusup, M.A. Kamaruddin, N. Khan, A Comprehensive review of crop yield prediction using machine learning approaches with special emphasis on palm oil yield prediction, IEEE Access 9 (2021) 63406–63439, doi:10.1109/ACCESS.2021.3075159.
[4] W. Yang, H. Feng, X. Hu, J. Song, J. Guo, B. Lu, An overview of high-throughput crop phenotyping: platform, image analysis, data mining, and data management, Plant Funct. Genomics: Methods Protocols 1 (2024) 3–8 Apr 25.
[5] J. Crossa, O.A. Montesinos-Lopez, G. Costa-Neto, P. Vitale, J.W. Martini, D. Runcie, R. Fritsche-Neto, A. Montesinos-Lopez, P. Pérez-Rodríguez, G. Gerard, S. Dreisigacker, Machine learning algorithms translate big data into predictive breeding accuracy, Trends Plant Sci. (2024) Oct 26.
[6] Z. Li, X. Zhou, Q. Cheng, W. Zhai, B. Mao, Y. Li, Z. Chen, An integrated feature selection approach to high water stress yield prediction, Front. Plant Sci. 14 (2023) 1289692 Dec 4.
[7] O.A. Montesinos-López, S. Ramos-Pulido, C.M. Hernandez-Suarez, B.A. Mosqueda González, F.A. Valladares-Anguiano, P. Vitale, A. Montesinos-López, J. Crossa, A novel method for genomic-enabled prediction of cultivars in new environments, Front. Plant Sci. 14 (2023) 1218151 Jul 25.
[8] S. Bhanumathi, M. Vineeth, N. Rohit, Crop Yield Prediction and Efficient use of Fertilizers, IEEE Xplore (2019) Apr. 01 https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8698087 .
[9] S. Lingwal, K.K. Bhatia, M. Singh, A novel machine learning approach for rice yield estimation, J. Exp. Theor. Artif. Intell. (2022) 1–20 Apr., doi:10.1080/0952813x.2022.2062458.
[10] S.S. Panda, D.P. Ames, S. Panigrahi, Application of vegetation indices for agricultural crop yield prediction using neural network techniques, Remote Sens (Basel) 2 (3) (2010) 673–696 Mar., doi:10.3390/rs2030673.
[11] N. Gnanasankaran, E. Ramaraj, T. Manikumar, An Intelligent framework for rice yield prediction using machine learning based models, Int. J. Sci. Eng. Res. 12 (1) (2021) [Online]. Available http://www.ijser.org .
[12] K.S. Saravanan, V. Bhagavathiappan, A comprehensive approach on predicting the crop yield using hybrid machine learning algorithms, J. Agrometeorol. 24 (2) (2022) 179–185 Jun., doi:10.54386/jam.v24i2.1561.
[13] A. Saxena, M. Dhadwal, M. Kowsigan, Indian crop production: prediction and model deployment using ML and streamlit, Turkish J. Physiotherapy and Rehabilitation 32 (3) (2021). [Online]. Available: www.turkjphysiotherrehabil.org.
[14] JCT College of Engineering and Technology, IEEE aerospace and electronic systems society, and institute of electrical and electronics engineers, ICISC 2018 : proceedings of the 2nd International Conference on Inventive Systems and Control (ICISC 2018): 19-20 January 2018.
[15] S. Kumar Sharma, D.P. Sharma, J.K. Verma, Study on machine-learning algorithms in crop yield predictions specific to Indian Agricultural Contexts, 2021 International Conference on Computational Performance Evaluation (ComPE), 2021 Dec., doi:10.1109/compe53109.2021.9752260.
[16] V. Sharma, M. Shukla, R. Mandal, Crop analysis and seed marketing using regression and association rules of India, International Conference on Emerging Trends in Information Technology and Engineering, IC-ETITE 2020, Institute of Electrical and Electronics Engineers Inc., 2020 Feb., doi:10.1109/ic-ETITE47903.2020.108.
[17] S.P. PG Scholar and P. J. Assistant Professor, "Crop yield prediction analysis using feed forward and recurrent neural network," 2020. [Online]. Available: www.ijisrt.com.
[18] IEEE Staff, 2019 Fifth International Conference on Image Information Processing (ICIIP). IEEE, 2019.
[19] B.M. Sagar, D.K. Cauvery, D. Rajkumar, Rice and wheat yield prediction in india using decision tree and random forest, Comput. Intell. Mach. Learn. 3 (2) (2022) 1–8.
[20] D. Elavarasan, P.M. Durairaj Vincent, Crop yield prediction using deep reinforcement learning model for sustainable agrarian applications, IEEE Access 8 (2020) 86886–86901, doi:10.1109/ACCESS.2020.2992480.
[21] L. Zhang, et al., Using boosted tree regression and artificial neural networks to forecast upland rice yield under climate change in the Sahel, Comput. Electron. Agric. 166 (2019) Nov., doi:10.1016/j.compag.2019.105031.
[22] J. Cao, et al., Integrating multi-source data for rice yield prediction across china using machine learning and deep learning approaches, Agric. For. Meteorol. 297 (2021) Feb., doi:10.1016/j.agrformet.2020.108275.
[23] S. Agarwal, S. Tarar, A hybrid approach for crop yield prediction using machine learning and deep learning algorithms, Journal of Physics: Conference Series, IOP Publishing Ltd, 2021 Jan., doi:10.1088/1742-6596/1714/1/012012.
[24] A. K. Mariappan and J. A. Ben Das, "A paradigm for rice yield prediction in Tamilnadu," 2017 IEEE Technological Innovations in ICT for Agriculture and Rural Development (TIAR), Chennai, India, 18-21 (2017) doi:10.1109/TIAR.2017.8273679.
[25] Z. Chu, J. Yu, An end-to-end model for rice yield prediction using deep learning fusion, Comput. Electron. Agric. 174 (2020) Jul., doi:10.1016/j.compag.2020.105471.

[26] S.S. Iyengar, V. Saxena, IEEE computer society. technical committee on parallel processing, institute of electrical and electronics engineers, Jaypee Institute of Information Technology University, and University of Florida. College of Engineering, 2019 Twelfth International Conference on Contemporary Computing (IC3-2019), Jaypee Institute of Information Technology, Noida, India, 2019 8-10 August.
[27] A. Chandra, P. Mitra, S.K. Dubey, S.S. Ray, Machine learning approach for kharif rice yield prediction integrating multi-temporal vegetation indices and weather and non-weather variables, in: International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives, International Society for Photogrammetry and Remote Sensing, 2019, pp. 187–194, doi:10.5194/isprs-archives-XLII-3-W6-187-2019.
[28] D.A. Bondre and M.S. Mahagaonkar, "Prediction of crop yield and fertilizer recommendation using machine learning algorithms," 2019. [Online]. Available: http://www.ijeast.com.
[29] P.N. Astya, Galgotias University. School of Computing Science and Engineering, Institute of Electrical and Electronics Engineers. Uttar Pradesh Section, and Institute of Electrical and Electronics Engineers, IEEE International Conference on Computing, Communication and Automation (ICCCA 2017): proceeding, 2017 on 5th-6th May.
[30] T. Hu, X. Zhang, G. Bohrer, Y. Liu, Y. Zhou, Y. Li, K. Zhao, Crop yield prediction via explainable AI and interpretable machine learning: dangers of black box models for evaluating climate change impacts on crop yield, Agric. For. Meteorol. 336 (2023) 109458, doi:10.1016/j.agrformet.2023.109458.
[31] Y. Goel, A. Vishwakarma, J. Malhotra, T. Gurram, K. Kadian, Enhancing Crop Yield Prediction Through Explainable AI for Interpretable Insights, in: A. Swaroop, V. Kansal, G. Fortino, A.E. Hassanien (Eds.), Proceedings of Fifth Doctoral Symposium on Computational Intelligence. DoSCI 2024. Lecture Notes in Networks and Systems, 1095, Springer, Singapore, 2024, doi:10.1007/978-981-97-6318-4_40.
[32] L. Breiman, Random Forests, Mach Learn 45 (1) (2001) 5–32, doi:10.1023/A:1010933404324.
[33] U. Kumar Dey, A.H. Masud, M.N. Uddin, Rice yield prediction model using data mining, in: 2017 International Conference on Electrical, Computer and Communication Engineering (ECCE), 2017, pp. 321–326. https://doi.org/10.1109/ecace.2017.7912925.
[34] P. Sathya, P. Gnanasekaran, Paddy yield prediction in tamil nadu delta region using MLR-LSTM Model, Appl. Artif. Intell. 37 (1) (2023), doi:10.1080/08839514.2023.2175113.
[35] E. Banu, A. Geetha, Rice crop yield prediction using random forest and deep neural network - An integrated approach, SSRN Electron. J. (2021), doi:10.2139/ssrn.3852547.
[36] S.K. Sharma, D.P. Sharma, K. Gaur, Crop yield predictions and recommendations using Random Forest Regression in 3A Agroclimatic zone, Rajasthan, J. Data Acquis. Proc. 38 (2) (2023) 1635, doi:10.5281/zenodo.776786.